



University
of Glasgow

TODO DATE AND TIME

Duration: 1.5 hours

Additional time: 30 minutes

Timed exam – fixed start time

DEGREE of MSc

Big Data: Systems, Prog & Mgt M COMPSCI 5088

(Answer all questions, note marks are not evenly distributed across questions)

This examination paper is an open book, online assessment and is worth a total of 60 marks.

1. (a) **Context:** Containers2You is a road-based hauling company that moves retail goods between warehouses around the country and across borders. Each truck used for hauling has a suite of sensors installed that send real-time data to company headquarters, including position, speed, direction and fuel consumption. The company currently operates 80 trucks, where a data packet is sent once each second as a text over a the local mobile phone data network. The company serves around 300 retailers across Europe.

Question: Currently this data is only used to produce company reports at the end of each month. However, Containers2You is considering allowing retailers using them to track where their goods are live, such that those retailers can be better prepared to unload their goods when the truck arrives. Would this change result in a Big Data problem?

Guidance: The question is worth 3 marks. 1 mark is awarded for correctly identifying whether this is a big data problem or not. 2 marks are awarded for reasons why (1 mark per point).

[3]

Solution:

- The answer is no [1 mark], as the company only needs to serve relatively small low-frequency data [1 mark] to a small number of businesses [1 mark].

- (b) **Context:** Over the past year Containers2You has been experiencing an increased number of insurance claims and truck damage due to poor road surfaces. They have installed dashboard cameras onto their trucks, and want to stream live video feeds back to headquarters via low-orbit satellite to monitor road conditions and then notify drivers to be careful or to re-route around dangerous areas. The camera's stream at 480p resolution, which translates to around 2MB of video data per minute, but the equipment quality was poor causing periodic loss of the video feeds.

Question: Processing the dashboard camera video constitutes a big data task, why?

Guidance: This question is worth 4 marks. We expect you to reference two dimensions that make this a big data task, along with why they are relevant to this scenario. 1 mark for a dimension and 2 marks for why.

[6]

Solution:

- Velocity [1 Mark]: The platform needs to process the images fast enough to alert drivers to issues with the road condition [2 Marks].
- Veracity [1 Mark]: The camera's have errors meaning that data cleaning will be needed to make the feeds usable [2 Marks].

Other explanations are acceptable, however, Volume and Variety are not valid conditions here.

- (c) **Context:** Management at Containers2You have become excited by the prospect of self-driving trucks. They believe that they might be able to use all of the sensor and video data

that they collect from their trucks to train an AI truck driver.

Question: What would be the legal and ethical implications of training such a model?

Guidance: This question is worth 6 marks. Three main points are expected, two marks per point.

[6]

Solution:

As a 'bad idea', there are lots of good answers to this, here are a sample, other reasonable arguments are acceptable:

- Privacy: The video cameras will also have recorded people's faces and car number-plates, using the raw footage would be an invasion of privacy [2 marks].
- Consent: The company did not get informed consent from people captured by the dashboard cameras [2 marks].
- Data Retention: EU legal requirements place limits on the amount of time a company can hold the videos, making model replacement in the future problematic. [2 marks]

2. (a) **Context:** UpBank is a multinational bank that provides personal and business banking, investment services, as well as cards and payments. Within the bank there is a data services division, which is responsible for data storage, database management and analytics services, which all other parts of the bank rely on. Day-to-day requests are handled by live databases that only hold recent customer and transaction data. However, much of the time-sensitive analytics work is performed on terabytes of historical data stored as raw text files in different formats such as CSV, as well as some special formats unique to the bank, where this data has been exported from the live databases on a daily-basis. The output of these analytic services are read into a custom legacy platform built by the bank for report generation.

Question: The data analytics team is considering implementing a standard interface for accessing these large stores of data, and are looking at a Hadoop-based solution for this. If they were to deploy Hadoop, what design decisions would they need to make to enable an effective deployment from a hardware and configuration perspective?

Guidance: This question is worth 6 marks. Three points are expected regarding Hadoop-specific deployment decisions. For each point there is one mark for what the design decision is and one mark for why this is important in the context of the above scenario.

[6]

Solution:

Any reasonable answers are acceptable here as long as they are specific to how Hadoop works, e.g. buying fast CPUs would not be a valid answer here. Examples of valid answers:

- **Distribution:** Hadoop is a distributed system, so the company will need to consider the number of compute nodes and hence the number of task-trackers needed to serve the different data-services users. [2 marks]
- **Storage:** Each machine would need sufficient local storage such that the aggregation of that storage is sufficient to hold all of the raw data [2 marks]
- **Co-Location:** The physical machines should be physically located close to each other to minimize the cost of network traffic, reducing execution time and cost. [2 Marks]
- **Redundant Job Trackers:** Multiple job trackers should be deployed to enable fall-over if one fails, since this is not supported naively and analytics is time sensitive. [2 Marks]

- (b) **Context:** Some of the data analytics team raise concerns regarding the amount of coding work that would be needed to replace the functionality of their current operations in Hadoop.

Question: Excluding the Map and Reduce functions as well as the main class that would need to be written, what other types of class might the development team need to implement?

Guidance: The question is worth 6 marks. three points are expected, two marks per point. For each point, 1 mark specifying the role that type of class fulfills and 1 mark is for why this is needed in the scenario above.

[6]

Solution:

This question is testing the students knowledge of the structure of a Map Reduce program:

- InputFormat: Hadoop needs a data reader that can convert the raw text files into records to be processed [1 Mark], where the bank has uniquely formatted data [1 Mark].
- OutputFormat: The output of a reduce needs to be written to disk via an output formatter [1 Mark]. As the bank needs to read data into a custom platform they will need to write the data in the format that platform expects [1 Mark].
- Object Classes: Classes that represent the intermediate analytics objects need to be implemented [1 Mark], as Hadoop follows an Object-Oriented Design and the bank data will be more complex than individual strings and numbers [1 Mark].

- (c) **Context:** A member of the trading team says that they need to compute full market statistics every 10 seconds, which is a List<Price> to <Double> transformation.

Question: Why might this be difficult/inefficient to perform in Hadoop?

Guidance: This question is worth 4 marks. Two main points are expected, for each, one mark is for a limitation in Hadoop and 1 mark is for why this is a problem for the above task.

[4]

Solution:

This is testing the student's knowledge of the limitations of the Hadoop API:

- The above task is a key-less transformation [1 Mark], where both Map and Reduce in Hadoop require keys [1 Mark].
- The computation that needs to be executed here needs to happen on a very regular basis [1 Mark], and Hadoop has significant start-up costs for jobs that would make this difficult [1 Mark].

Other similar points are acceptable.

3. (a) **Context:** MoonShuttle is a crypto-currency exchange and broker, where their business model relies on front-running the trades of their users (effectively imposing a hidden tax on transactions). To enable this, the exchange needs to hold customer, account and transaction data in a similar way to a normal bank, but also need very low-latency look-up of incoming trade requests to make a profit from each transaction.

Question: MoonShuttle currently uses an HBase solution to store incoming trade data. Summarize the storage mechanism that HBase Region Servers use and describe how this enables fast access for recent trades.

Guidance: This question is worth 7 marks. 1 mark is for naming the storage mechanism. 4 marks are for providing a bullet point summary of the write steps that a trade will go through, one mark per step. 2 marks are awarded for summarizing why this allows for fast look-up of recent trades.

[7]

Solution:

This question is asking for a summary of how a Log-Structured Merge (LSM) Tree [1 Mark] works in HBase:

- Writes the trade into a Write Ahead Log, appending to the file [1 Mark]
- Also stores the trade in a MemTable [1 Mark]
- Periodically the trades will get sorted, compressed and written into an SSTable on disk [1 Mark].
- SSTables are compacted periodically to save storage space [1 Mark].

The reason this allows for fast look-up of trades is that recent trade data will be stored in the MemTable, which can be directly queried [1 Mark]. This acts like a recent cache that can be accessed at main-memory speeds [1 Mark].

- (b) **Context:** MoonShuttle currently has their HBase installation configured such that it writes its tables to disk infrequently, resulting in a small number of large files being written. However, the MoonShuttle analytics team report that look-up of older transaction records is slow, and this is traced to the loading of the large files into memory prior to being scanned for the target records.

Question: One of the engineers says that they store few large files since it reduces the index look-up time required to find the table file that has the target records. Is this true and why?

Guidance: This question is worth 3 marks. 1 mark is available for whether the statement is true, while 2 marks are available for why.

[3]

Solution:

This is not true [1 Mark]. HBase uses Bloom filters to facilitate SSTable file index look-up [1 Mark]. This is very fast as it has $O(k)$ complexity, where k is the maximum

number of hash functions, which can be computed in parallel leading to a constant time index lookup [1 Mark].

- (c) **Context:** MoonShuttle executives are worried that a government in one region that they operate might demand access to all of their customer records, which are all stored in a single HBase instance. One of their team suggests transitioning to a solution based on Apache Cassandra.

Question: What benefits might a Cassandra-based solution offer over the current HBase installation for MoonShuttle?

Guidance: This question is worth 4 marks. 2 points are expected, for each 1 mark is for the relevant feature of Cassandra and 1 mark is for why this helps alleviate the executive's concerns.

[4]

Solution:

- Cassandra has no central master index of records and no scan functionality in its API [1 Mark] meaning that it would be difficult to obtain a full list of MoonShuttle customers [1 Mark].
- Cassandra uses a hashing scheme to select which arc data is stored on [1 Mark], this could be used to geographically partition user data by country while maintaining searchability [1 Mark].

Other differences such as how search works (e.g. via DHT overlays) are not relevant here.

- (d) **Context:** MoonShuttle decides to migrate to Cassandra and starts deploying Cassandra nodes to different geographical regions, however they find this to be difficult and time-consuming to deploy across the heterogeneous hardware available in different regions and also problematic to maintain due to node failures.

Question: What is Kubernetes and how might it help here?

Guidance: This question is worth 5 marks. 2 marks are for defining Kubernetes, while three marks are for how this might help, 1 mark per point.

[5]

Solution:

- Kubernetes is an open-source container orchestration system [1 Mark] for automating software deployment, scaling, and management [1 Mark].
- Kubernetes would allow MoonShuttle to automate the deployment of Cassandra nodes when packaged as containers [1 Mark].
- Kubernetes can automatically restart failed Cassandra nodes [1 Mark].

- Kubernetes abstracts away from the physical hardware, allowing for a unified deployment pipeline [1 Mark].

4. (a) **Context:** WhiskyAI is a start-up specializing in the training of foundational artificial intelligence models (like ChatGPT or DeepSeek use). To reduce training costs, WhiskyAI uses large farms of cheap desktop-class machines to train their models. At each training step, all machines are given a different batch of training examples to learn from, which update a local neural network on each machine. After every 10 training steps, the neural networks from all machines need to be synchronized (brought together and combined into a single model), and then redistributed before the next training step.

Question: Why would this training process be more costly and difficult to efficiently execute in a distributed big data framework like Spark than applications that don't produce side effects that need periodic synchronization?

Guidance: This question is worth 6 marks. 2 marks are for stating what type of big data problem this is, while 4 marks are for providing reasons why this difficult or costly - two points are expected, 2 marks per point.

[6]

Solution:

- The core problem is that this training process requires global state, i.e. the application is stateful [2 Marks]
- Implementing (globally) stateful distributed processing is difficult since:
 - The synchronization process of the state adds an intermediate step to the process, where the cluster machines are waiting [2 Marks]
 - Each transformation task has to store this additional state (in GPU memory in this case), which over time can cause out-of-memory errors as this footprint grows [2 Marks]
 - * (this technically should not happen for the above scenario as a NN should be constant size, but I cannot expect that the students will know that, so this is an acceptable answer)
 - The intermediate state needs to be transferred across the network that adds additional cost [2 marks].
 - Error recovery is more complex as a worker/executor failure also loses the associated state, meaning the entire batch needs to be re-executed [2 marks].

Other reasonable discussions on why this is difficult are acceptable, 2 marks per point.

- (b) **Context:** One of the issues that WhiskyAI observes is that performing learning in batches wastes compute resources, as each training step involves first preparing a batch (where the CPU is working and the GPU is idle) and the neural network updating (where the GPU is working and the CPU is idle).

Question: One of the WhiskyAI development team proposes switching to a streaming-based pipeline instead. Why might this result in better compute resource usage?

Guidance: This question is worth 4 marks. 2 marks are for what would make a streaming

solution different, while 2 marks are for why this solves the above problem.

[4]

Solution:

- Stream-based solutions define computation as a pipeline processing steps, where items can be in different stages of the pipeline can be processed in parallel [2 Marks]. Assuming the implementation separated batch and network updates as different pipeline steps, the next batch can be getting prepared while learning from the current batch proceeds [2 Marks].

Other reasonable discussions on why this is difficult are acceptable, 2 marks per point.